

# StyCLIP: StyleTransfer in Few-Shot Learning

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**Abstract**—Few-shot learning for visual artist classification poses a unique challenge due to the diversity of artistic styles and limited labeled data per class. As a baseline, we evaluate the performance of various CLIP-based models on this task. To enhance model generalization, we explore the integration of style transfer techniques directly into the CLIP architecture, hypothesizing that stylization-aware features may better capture artist-specific patterns. However, experimental results on a curated painting dataset show that this approach consistently underperforms compared to standard CLIP variants. Our findings suggest that incorporating style transfer into the model architecture does not yield meaningful gains in artist classification performance under few-shot conditions. The code is available at <https://github.com/Fonty02/ComputerVision/tree/main/TIMO>.

## I. INTRODUCTION

Visual artist classification represents a fundamental challenge in computer vision, bridging art history and automated content analysis. The exponential growth of digitized art collections necessitates automated systems to reduce manual effort while democratizing access to art historical knowledge.

We address few-shot learning for artist classification, where models must identify artistic styles using limited labeled examples. This task is particularly challenging as artistic style encompasses subtle visual characteristics like brushstroke patterns, color palettes, and compositional techniques requiring specialized domain knowledge.

While CLIP-based few-shot methods excel at semantic understanding, they may overlook fine-grained stylistic nuances crucial for artist identification. We propose StyCLIP, a novel CLIP extension incorporating artistic style through Gram matrix features. Our contributions include: (1) a StyleAdapter module integrating style-aware features with semantic representations, (2) comprehensive evaluation using a curated ArtGraph dataset subset, and (3) extensive experiments to evaluate the impact of style-aware features on few-shot artist classification.

## II. RELATED WORK

### A. Style Transfer

Style transfer techniques, pioneered by Gatys et al. [1], revealed that deep neural networks implicitly encode stylistic information in their intermediate layers. Their seminal work showed that the correlations between feature activations, captured through Gram matrices, effectively represent artistic style independently of content.

The fundamental insight behind neural style transfer lies in the separation of content and style representations within convolutional neural networks. Content information, which describes the structural and semantic aspects of an image, is primarily captured by the spatial arrangement of feature activations in deeper layers. Style information, representing textural patterns, color distributions, and artistic characteristics, is encoded in the correlations between feature maps across different channels.

Gram matrices serve as the mathematical foundation for capturing these style correlations. Given a feature map  $F^l \in \mathbb{R}^{C_l \times H_l \times W_l}$  from layer  $l$  with  $C_l$  channels and spatial dimensions  $H_l \times W_l$ , the Gram matrix  $G^l \in \mathbb{R}^{C_l \times C_l}$  is computed as:

$$G_{i,j}^l = \sum_{k=1}^{H_l} \sum_{m=1}^{W_l} F_{i,k,m}^l \cdot F_{j,k,m}^l$$

This computation measures the inner product between vectorized feature maps of channels  $i$  and  $j$ , effectively capturing how often different visual patterns co-occur spatially. The resulting Gram matrix encodes second-order statistics that are invariant to spatial location but sensitive to the distribution and correlation of visual features.

The normalization of Gram matrices is typically performed by dividing by the total number of spatial locations ( $H_l \times W_l$ ) to ensure scale invariance:

$$G_{i,j}^l = \frac{1}{H_l \times W_l} \sum_{k=1}^{H_l} \sum_{m=1}^{W_l} F_{i,k,m}^l \cdot F_{j,k,m}^l$$

This normalization allows style representations to be independent of image size and enables consistent style transfer across images of different resolutions.

While originally developed for style transfer applications, Gram matrix representations offer valuable insights into modeling artistic style for classification tasks. The ability to capture style information independently of content makes them particularly suitable for artist identification, where stylistic consistency across diverse subjects is a key discriminative factor.

### B. Few-Shot Learning and Meta-Learning

Few-shot learning (FSL) methodologies aim to develop models that can generalize from limited examples, making them particularly relevant for artistic domains where comprehensive datasets are rare. Meta-learning approaches Prototypical Networks [2] fundamentally changed the FSL landscape by learning to learn from small samples.

### C. Contrastive Language-Image Pretraining

The emergence of large-scale vision-language models like CLIP [3] has revolutionized few-shot artistic recognition by leveraging textual descriptions of artistic styles. CLIP learns multimodal representations by maximizing cosine similarity between correct image-text pairs while minimizing similarity for incorrect pairs through contrastive learning.

The CLIP architecture consists of visual and text encoders that project inputs into a shared embedding space. The visual encoder can be Vision Transformer (ViT) or ResNet-based (like RN50), while the text encoder uses Transformer architecture with self-attention mechanisms. Both encoders map to a common D-dimensional space (typically 512 or 1024 dimensions), enabling zero-shot classification through embedding similarity comparison.

Methods like TIP-Adapter [4] incorporate few-shot examples through a non-parametric cache model that complements zero-shot predictions. Other approaches like APE [5] enhance performance by selectively refining CLIP’s features based on adaptive prior refinement, while GDA-CLIP [6] introduces a novel gradient-based adaptation technique to optimize decision boundaries. More sophisticated approaches like TIMO [7] leverage mutual guidance between text and image modalities to improve low-shot performance.

Our work investigates integrating Gram matrix-based style representations into CLIP architectures to enhance few-shot artist classification performance.

### III. DATA

We utilize a subset of the ArtGraph dataset [8], a comprehensive art knowledge graph containing over 100,000 paintings with metadata. We created two non-overlapping subsets: one for few-shot evaluation and another for StyCLIP training.

#### A. ArtGraph Few-Shot Subset

Our few-shot dataset follows a systematic two-stage selection process:

- **Style Selection:** Top 10 artistic styles by artwork count
- **Artist Filtering:** Artists with 30-50 artworks per style (up to 10 artists per style)

**Data Splits:** 80% support set, 10% validation, 10% query set to accommodate varying portfolio sizes.

#### B. ArtGraph StyCLIP Subset

For style-aware training, we use complementary artists (16-50 artworks) excluded from the few-shot subset, ensuring no data leakage between training and evaluation phases.

### IV. EXPERIMENTS

In this section, we describe the experimental setup used to create, train, StyCLIP, and evaluate the models. All these experiments were conducted on a single NVIDIA RTX 3060 GPU with 12GB of memory, 16GB of 3200MHz DDR4 RAM and Ryzen 5 3600 CPU.

#### A. StyCLIP structure

StyCLIP extends the standard CLIP architecture by integrating explicit style representations through Gram matrix features, specifically designed to capture artistic style information that complements semantic understanding. This section details the architectural components and modifications introduced in our approach.

1) *Standard CLIP with RN50 Backbone:* The baseline CLIP model with ResNet-50 backbone consists of:

- **Visual Encoder:** To extract visual features from images into a 1024-dimensional embedding space
- **Text Encoder:** To extract semantic features from text descriptions into a 1024-dimensional embedding space
- **Contrastive Learning:** To align visual and text embeddings in a shared space using a contrastive loss function

2) *StyCLIP Architecture Overview:* Our StyCLIP architecture maintains the original CLIP backbone while introducing three key components for style-aware feature learning:

- **Gram Matrix Extraction Module:** Extracts style-specific correlations from intermediate visual features of the ResNet-50 backbone to capture stylistic patterns at different resolution scales. The resulting matrices represent channel correlations that encode texture information, brushstrokes, and artistic characteristics independent of semantic content.
- **Style Feature Projection Layers:** Perform dimensionality reduction of high-dimensional Gram matrix representations. They use a Multi-Layer Perceptron (MLP) to project concatenated Gram matrices (after extracting the upper triangular part for computational efficiency) from a high-dimensional space to a compact 256-dimensional space, preserving the most relevant stylistic information for artist classification.
- **Fusion Adapter:** Integrates CLIP semantic features and stylistic features through a bottleneck architecture. The module concatenates CLIP visual features (512 dimensions) with projected stylistic features (256 dimensions) and uses a bottleneck MLP adapter that reduces dimensionality (768 bottleneck dimension hyperparameter 512) to effectively fuse semantic and stylistic information into a unified representation optimized for few-shot artist classification.

3) *Training Strategy:* During training, the CLIP backbone remains frozen while only the style projection layers and fusion adapter are optimized. This approach:

- Preserves pre-trained semantic knowledge from CLIP
- Reduces computational requirements by limiting trainable parameters
- Enables efficient fine-tuning on artistic datasets
- Maintains stability in the learned representations

#### B. StyCLIP hyperparameters tuning and training

StyCLIP hyperparameters are optimized using Optuna [9] with the following search space:

**Architectural Parameters:** Fusion bottleneck dimension {64, 128, 256}; dropout rate [0.0, 0.5, step=0.1]

**Training Parameters:** Learning rate log-uniform [ $10^{-5}$ ,  $10^{-3}$ ]; weight decay log-uniform [ $10^{-6}$ ,  $10^{-3}$ ]

1) *Meta-Learning Training Protocol:* Our training employs a prototypical network [2] approach adapted for artistic classification:

**Episodic Training Structure:** Each training episode consists of N-way classification tasks where N=10 artists are randomly sampled. For each artist, we sample K support examples and Q=3 query examples, creating balanced few-shot scenarios.

**K-Shot Experimental Design:** Our evaluation examines StyCLIP performance across two key support set configurations to understand how training data quantity affects style-aware few-shot learning:

- **1-Shot (K=1):** Minimal data scenario testing generalization from single examples per artist
- **4-Shot (K=4):** Enhanced data setting enabling more robust prototype estimation and style variation capture

This comparison reveals how explicit style modeling benefits vary with available training examples.

2) *Training Configuration:* We employ AdamW optimization with cosine annealing learning rate scheduling. A warmup period of 100 steps ensures stable training initialization. Training incorporates early stopping with patience=10 epochs and minimum improvement threshold of 0.001 to prevent overfitting while maintaining training efficiency. For robust hyperparameter selection, we use a separate meta-validation set comprising 30% of available artists, ensuring no overlap with training data. Only the newly introduced components (Gram matrix projections and fusion adapter) are trainable.

3) *Optimization Strategy:* The Optuna framework enables efficient hyperparameter exploration using Tree-structured Parzen Estimator (TPE) for intelligent sampling, median pruning for early termination of unpromising trials, and 15 trials per K-shot configuration to balance exploration with computational efficiency.

This systematic approach ensures robust hyperparameter selection while maintaining computational feasibility for the K-shot comparative analysis.

#### C. Evaluation of different models

We conduct comprehensive experiments to evaluate StyCLIP against established few-shot learning baselines using a systematic evaluation protocol designed to ensure robust and statistically significant results.

1) *Experimental Protocol:* Our evaluation follows a standard few-shot learning protocol where each test episode represents an N-way classification task. For our artist classification domain:

- **N-way Classification:** Each episode involves classifying among all available artist classes in our curated dataset subset
- **Support Set:** K examples per artist class are used to form class prototypes
- **Query Set:** Remaining examples serve as query instances for classification evaluation

We evaluate model performance across multiple shot settings to understand how the number of support examples affects classification accuracy: K {1, 2, 4, 8, 16} shots per class

2) *Statistical Robustness:* To ensure statistical validity and account for variance in few-shot performance, we implement a comprehensive multi-seed evaluation strategy:

Each experimental configuration is evaluated across 10 different random seeds: {1, 2, 3, 4, 5, 6, 7, 8, 9, 42}. This approach controls for:

- **Support Set Sampling Variance:** Different random selections of K examples per class
- **Model Initialization Effects:** Stochastic components in model training and adaptation
- **Episode Construction Randomness:** Variation in query set composition across evaluation runs

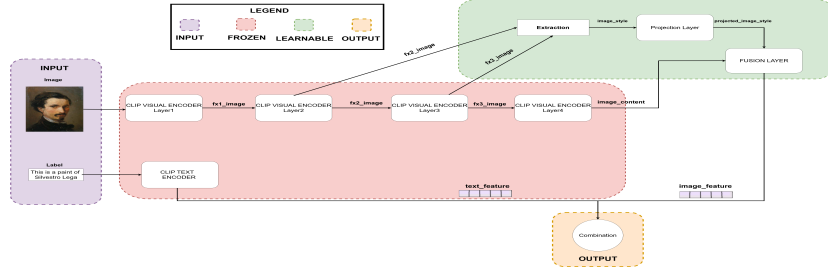


Figure 1: StyCLIP Architecture Overview

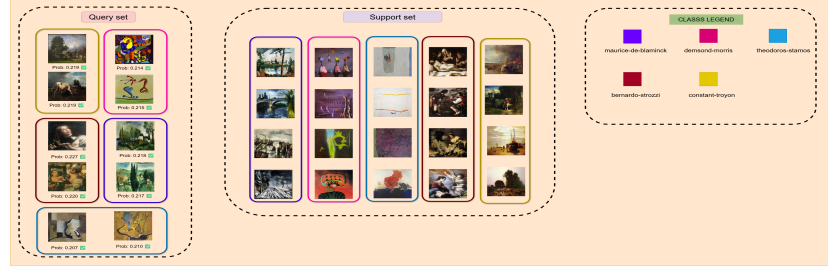


Figure 2: Example of and 10 Way 4-Shot Episode

3) *Baseline Methods*: We perform several experiments by changing the model used for the few-shot learning task and the backbone used for the CLIP model. The following models are used:

- **CLIP + TIP-Adapter**: Standard CLIP model with TIP-Adapter [4] for few-shot adaptation
- **CLIP + APE**: Standard CLIP model with Adaptive Prior Refinement [5] for few-shot adaptation
- **CLIP + GDA-CLIP**: Standard CLIP model with Gradient-based Domain Adaptation [6] for few-shot adaptation
- **CLIP + TIMO**: Standard CLIP model with TIMO [7] for few-shot adaptation
- **CLIP + TIMO-S**: Our proposed StyCLIP model with integrated style-aware features

with the following backbones:

- **CLIP RN50**: Standard CLIP model with ResNet-50 backbone
- **StyCLIP FSL1**: StyCLIP model with ResNet-50 backbone trained with 1-shot configuration
- **StyCLIP FSL4**: StyCLIP model with ResNet-50 backbone trained with 4-shot configuration

4) *Evaluation Metrics*: Performance assessment employs multiple complementary metrics to provide comprehensive evaluation: **Accuracy**, **Macro-averaged Precision**, **Macro-averaged Recall**, and **Macro-averaged F1-Score**.

## B. Evaluation Results

The following tables present detailed performance metrics for all tested model-backbone combinations across 1-shot, 2-shot, 4-shot, 8-shot, and 16-shot scenarios. Here each results has been multiplied by 100 to express them as percentages.

Table I: 1-Shot Classification Results

| Model       | Backbone     | ACC          | Precision    | Recall       | F1           |
|-------------|--------------|--------------|--------------|--------------|--------------|
| APE         | StyCLIP_FSL1 | 2.05         | 0.93         | 2.05         | 0.87         |
| APE         | StyCLIP_FSL4 | 2.48         | 0.81         | 2.44         | 0.84         |
| APE         | RN50         | 44.31        | 44.42        | 43.94        | 39.76        |
| GDA_CLIP    | StyCLIP_FSL1 | 13.49        | 13.88        | 13.23        | 11.98        |
| GDA_CLIP    | StyCLIP_FSL4 | 16.33        | 17.48        | 16.08        | 14.62        |
| GDA_CLIP    | RN50         | 41.41        | 38.86        | 40.76        | 35.33        |
| TIMO        | StyCLIP_FSL1 | 6.97         | 6.88         | 6.81         | 6.13         |
| TIMO        | StyCLIP_FSL4 | 7.37         | 7.57         | 7.17         | 6.69         |
| TIMO        | RN50         | 43.88        | 44.32        | 43.52        | 40.29        |
| TIMO_S      | StyCLIP_FSL1 | 7.28         | 7.27         | 7.11         | 6.38         |
| TIMO_S      | StyCLIP_FSL4 | 7.31         | 7.56         | 7.13         | 6.63         |
| TIMO_S      | RN50         | <b>45.96</b> | <b>44.47</b> | <b>45.54</b> | <b>41.07</b> |
| Tip_Adapter | StyCLIP_FSL1 | 1.31         | 0.03         | 1.34         | 0.06         |
| Tip_Adapter | StyCLIP_FSL4 | 1.13         | 0.01         | 1.02         | 0.03         |
| Tip_Adapter | RN50         | 38.65        | 30.85        | 38.38        | 30.92        |

## V. RESULTS

### A. Hyperparameter Optimization Results

The Optuna-based hyperparameter optimization process identified the optimal configurations for the two StyCLIP training scenarios.

For the FSL1 (1-shot) model, the best hyperparameters are:

- Fusion Bottleneck Dimension: 64
- Learning Rate:  $6.01 \times 10^{-5}$
- Dropout Rate: 0.0
- Weight Decay:  $4.83 \times 10^{-4}$
- Validation Accuracy: 57.07%

For the FSL4 (4-shot) model, the best hyperparameters are:

- Fusion Bottleneck Dimension: 128
- Learning Rate:  $1.02 \times 10^{-5}$
- Dropout Rate: 0.1
- Weight Decay:  $1.03 \times 10^{-6}$
- Validation Accuracy: 60.53%

Table II: 2-Shot Classification Results

| Model       | Backbone     | ACC          | Precision    | Recall       | F1           |
|-------------|--------------|--------------|--------------|--------------|--------------|
| APE         | StyCLIP_FSL1 | 3.55         | 1.51         | 3.51         | 1.51         |
| APE         | StyCLIP_FSL4 | 3.39         | 2.03         | 3.38         | 1.71         |
| APE         | RN50         | 47.09        | 47.77        | 46.65        | 42.37        |
| GDA_CLIP    | StyCLIP_FSL1 | 20.95        | 21.79        | 20.59        | 19.40        |
| GDA_CLIP    | StyCLIP_FSL4 | 25.96        | 27.35        | 25.48        | 24.22        |
| GDA_CLIP    | RN50         | 47.22        | 47.90        | 46.41        | 42.47        |
| TIMO        | StyCLIP_FSL1 | 12.14        | 11.82        | 11.87        | 10.78        |
| TIMO        | StyCLIP_FSL4 | 13.55        | 13.67        | 13.21        | 12.16        |
| TIMO        | RN50         | 47.89        | 48.43        | 47.32        | 43.72        |
| TIMO_S      | StyCLIP_FSL1 | 13.76        | 13.84        | 13.45        | 12.44        |
| TIMO_S      | StyCLIP_FSL4 | 15.38        | 15.69        | 15.02        | 13.92        |
| TIMO_S      | RN50         | <b>48.65</b> | <b>48.78</b> | <b>47.95</b> | <b>43.98</b> |
| Tip_Adapter | StyCLIP_FSL1 | 1.19         | 0.02         | 1.19         | 0.04         |
| Tip_Adapter | StyCLIP_FSL4 | 1.19         | 0.02         | 1.08         | 0.03         |
| Tip_Adapter | RN50         | 40.73        | 32.99        | 40.33        | 32.58        |

Table III: 4-Shot Classification Results

| Model       | Backbone     | ACC          | Precision    | Recall       | F1           |
|-------------|--------------|--------------|--------------|--------------|--------------|
| APE         | StyCLIP_FSL1 | 4.62         | 2.09         | 4.42         | 2.17         |
| APE         | StyCLIP_FSL4 | 5.29         | 2.75         | 5.12         | 2.62         |
| APE         | RN50         | 53.88        | 54.93        | 53.72        | 49.73        |
| GDA_CLIP    | StyCLIP_FSL1 | 27.16        | 28.56        | 26.84        | 25.48        |
| GDA_CLIP    | StyCLIP_FSL4 | 35.17        | 38.35        | 34.79        | 33.74        |
| GDA_CLIP    | RN50         | 55.54        | 58.24        | 54.97        | 52.66        |
| TIMO        | StyCLIP_FSL1 | 21.77        | 21.38        | 21.35        | 19.75        |
| TIMO        | StyCLIP_FSL4 | 23.43        | 25.00        | 23.06        | 22.13        |
| TIMO        | RN50         | 56.48        | 59.14        | 55.90        | 53.96        |
| TIMO_S      | StyCLIP_FSL1 | 23.24        | 23.04        | 22.81        | 21.19        |
| TIMO_S      | StyCLIP_FSL4 | 26.36        | 28.69        | 25.97        | 25.27        |
| TIMO_S      | RN50         | <b>57.28</b> | <b>59.77</b> | <b>56.74</b> | <b>54.55</b> |
| Tip_Adapter | StyCLIP_FSL1 | 1.38         | 0.04         | 1.40         | 0.08         |
| Tip_Adapter | StyCLIP_FSL4 | 1.28         | 0.02         | 1.19         | 0.04         |
| Tip_Adapter | RN50         | 44.95        | 39.16        | 44.38        | 37.19        |

Table IV: 8-Shot Classification Results

| Model       | Backbone     | ACC          | Precision    | Recall       | F1           |
|-------------|--------------|--------------|--------------|--------------|--------------|
| APE         | StyCLIP_FSL1 | 5.35         | 2.57         | 5.27         | 2.57         |
| APE         | StyCLIP_FSL4 | 7.52         | 4.71         | 7.35         | 4.42         |
| APE         | RN50         | 58.10        | 58.94        | 57.97        | 54.03        |
| GDA_CLIP    | StyCLIP_FSL1 | 34.25        | 35.76        | 33.93        | 32.55        |
| GDA_CLIP    | StyCLIP_FSL4 | 46.97        | 50.15        | 46.42        | 45.55        |
| GDA_CLIP    | RN50         | 63.61        | 67.18        | 63.19        | 61.57        |
| TIMO        | StyCLIP_FSL1 | 30.64        | 30.61        | 30.49        | 28.71        |
| TIMO        | StyCLIP_FSL4 | 38.38        | 40.46        | 37.90        | 36.52        |
| TIMO        | RN50         | <b>65.26</b> | <b>68.43</b> | <b>64.93</b> | <b>63.66</b> |
| TIMO_S      | StyCLIP_FSL1 | 32.17        | 33.14        | 31.92        | 30.49        |
| TIMO_S      | StyCLIP_FSL4 | 40.89        | 43.57        | 40.41        | 39.13        |
| TIMO_S      | RN50         | 65.08        | 68.25        | 64.75        | 63.45        |
| Tip_Adapter | StyCLIP_FSL1 | 1.71         | 0.21         | 1.72         | 0.24         |
| Tip_Adapter | StyCLIP_FSL4 | 1.38         | 0.03         | 1.28         | 0.06         |
| Tip_Adapter | RN50         | 48.44        | 45.26        | 47.99        | 41.20        |

Table V: 16-Shot Classification Results

| Model       | Backbone     | ACC          | Precision    | Recall       | F1           |
|-------------|--------------|--------------|--------------|--------------|--------------|
| APE         | StyCLIP_FSL1 | 6.73         | 3.35         | 6.55         | 3.37         |
| APE         | StyCLIP_FSL4 | 7.98         | 5.63         | 7.87         | 5.13         |
| APE         | RN50         | 62.17        | 63.65        | 61.95        | 58.36        |
| GDA_CLIP    | StyCLIP_FSL1 | 41.19        | 42.45        | 41.00        | 39.13        |
| GDA_CLIP    | StyCLIP_FSL4 | 53.36        | 55.25        | 52.86        | 51.64        |
| GDA_CLIP    | RN50         | 73.06        | 76.12        | 72.58        | 71.63        |
| TIMO        | StyCLIP_FSL1 | 37.00        | 37.83        | 36.87        | 34.90        |
| TIMO        | StyCLIP_FSL4 | 43.27        | 45.16        | 42.80        | 41.36        |
| TIMO        | RN50         | 73.36        | 76.59        | 72.94        | 72.09        |
| TIMO_S      | StyCLIP_FSL1 | 39.20        | 40.52        | 39.01        | 37.22        |
| TIMO_S      | StyCLIP_FSL4 | 48.75        | 50.58        | 48.27        | 46.90        |
| TIMO_S      | RN50         | <b>73.46</b> | <b>76.52</b> | <b>73.04</b> | <b>72.10</b> |
| Tip_Adapter | StyCLIP_FSL1 | 2.63         | 0.70         | 2.60         | 0.74         |
| Tip_Adapter | StyCLIP_FSL4 | 1.77         | 0.07         | 1.69         | 0.13         |
| Tip_Adapter | RN50         | 52.91        | 51.90        | 52.37        | 46.38        |

### C. Statistical Analysis

Table VI: Statistical Comparison Matrix for 1-Shot Classification (Accuracy %). \* indicates significant differences ( $p < 0.05$ ) between models using t-test.

| Model       | APE_FSL1 | APE_FSL4 | APE_RN50 | GDA_FSL1 | GDA_FSL4 | GDA_RN50 | TIMO_FSL1 | TIMO_FSL4 | TIMO_RN50 | TIMO_S_FSL1 | TIMO_S_FSL4 | TIMO_S_RN50 | Tip_FSL1 | Tip_FSL4 | Tip_RN50 |
|-------------|----------|----------|----------|----------|----------|----------|-----------|-----------|-----------|-------------|-------------|-------------|----------|----------|----------|
| APE_FSL1    | 0.00     | -0.43    | -42.26*  | -11.44*  | -14.28*  | -39.36*  | -4.92*    | -5.32*    | -41.83*   | -5.23*      | -5.26*      | -43.91*     | 0.73     | 0.92*    | -36.61*  |
| APE_FSL4    | 0.43     | 0.00     | -41.83*  | -11.01*  | -13.85*  | -38.93*  | -4.50*    | -4.89*    | -41.41*   | -4.80*      | -4.83*      | -43.49*     | 1.16*    | 1.35*    | -36.18*  |
| APE_RN50    | 42.26*   | 41.83*   | 0.00     | 30.83*   | 27.98*   | 2.91*    | 37.34*    | 36.94*    | 0.43      | 37.03*      | 37.00*      | -1.65       | 43.00*   | 43.18*   | 5.66*    |
| GDA_FSL1    | 11.44*   | 11.01*   | -30.83*  | 0.00     | -2.84*   | -27.92*  | 6.51*     | 6.12*     | -30.40*   | 6.21*       | 6.18*       | -32.48*     | 12.17*   | 12.35*   | -25.17*  |
| GDA_FSL4    | 14.28*   | 13.85*   | -27.98*  | 2.84*    | 0.00     | -25.08*  | 9.36*     | 8.96*     | -27.55*   | 9.05*       | 9.02*       | -29.63*     | 15.02*   | 15.20*   | -22.32*  |
| GDA_RN50    | 39.36*   | 38.93*   | -2.91*   | 27.92*   | 25.08*   | 0.00     | 34.43*    | 34.04*    | -2.48*    | 34.13*      | 34.10*      | -4.56*      | 40.09*   | 40.28*   | 2.75*    |
| TIMO_FSL1   | 4.92*    | 4.50*    | -37.34*  | -6.51*   | -9.36*   | -34.43*  | 0.00      | -0.40     | -36.91*   | -0.31       | -0.34       | -38.99*     | 5.66*    | 5.84*    | -31.68*  |
| TIMO_FSL4   | 5.32*    | 4.89*    | -36.94*  | -6.12*   | -8.96*   | -34.04*  | 0.40      | 0.00      | -36.51*   | 0.09        | 0.06        | -38.59*     | 6.06*    | 6.24*    | -31.28*  |
| TIMO_RN50   | 41.83*   | 41.41*   | -0.43    | 30.40*   | 27.55*   | 2.48*    | 36.91*    | 36.51*    | 0.00      | 36.61*      | 36.57*      | -2.08       | 42.57*   | 42.75*   | 5.23*    |
| TIMO_S_FSL1 | 5.23*    | 4.80*    | -37.03*  | -6.21*   | -9.05*   | -34.13*  | 0.31      | -0.09     | -36.61*   | 0.00        | -0.03       | -38.69*     | 5.96*    | 6.15*    | -31.38*  |
| TIMO_S_FSL4 | 5.26*    | 4.83*    | -37.00*  | -6.18*   | -9.02*   | -34.10*  | 0.34      | -0.06     | -36.57*   | 0.03        | 0.00        | -38.65*     | 5.99*    | 6.18*    | -31.35*  |
| TIMO_S_RN50 | 43.91*   | 43.49*   | 1.65     | 32.48*   | 29.63*   | 4.56*    | 38.99*    | 38.59*    | 2.08      | 38.69*      | 38.65*      | 0.00        | 44.65*   | 44.83*   | 7.31*    |
| Tip_FSL1    | -0.73    | -1.16*   | -43.00*  | -12.17*  | -15.02*  | -40.09*  | -5.66*    | -6.06*    | -42.57*   | -5.96*      | -5.99*      | -44.65*     | 0.00     | 0.18     | -37.34*  |
| Tip_FSL4    | -0.92*   | -1.35*   | -43.18*  | -12.35*  | -15.20*  | -40.28*  | -5.84*    | -6.24*    | -42.75*   | -6.15*      | -6.18*      | -44.83*     | -0.18    | 0.00     | -37.52*  |
| Tip_RN50    | 36.61*   | 36.18*   | -5.66*   | 25.17*   | 22.32*   | -2.75*   | 31.68*    | 31.28*    | -5.23*    | 31.38*      | 31.35*      | -7.31*      | 37.34*   | 37.52*   | 0.00     |

Table VII: Statistical Comparison Matrix for 2-Shot Classification (Accuracy %). \* indicates significant differences ( $p < 0.05$ ) between models using t-test.

| Model       | APE_FSL1 | APE_FSL4 | APE_RN50 | GDA_FSL1 | GDA_FSL4 | GDA_RN50 | TIMO_FSL1 | TIMO_FSL4 | TIMO_RN50 | TIMO_S_FSL1 | TIMO_S_FSL4 | TIMO_S_RN50 | Tip_FSL1 | Tip_FSL4 | Tip_RN50 |
|-------------|----------|----------|----------|----------|----------|----------|-----------|-----------|-----------|-------------|-------------|-------------|----------|----------|----------|
| APE_FSL1    | 0.00     | 0.15     | -43.55*  | -17.40*  | -22.42*  | -43.67*  | -8.59*    | -10.00*   | -44.34*   | -10.21*     | -11.83*     | -45.11*     | 2.35*    | 2.35*    | -37.19*  |
| APE_FSL4    | -0.15    | 0.00     | -43.70*  | -17.55*  | -22.57*  | -43.82*  | -8.75*    | -10.15*   | -44.50*   | -10.37*     | -11.99*     | -45.26*     | 2.20*    | 2.20*    | -37.34*  |
| APE_RN50    | 43.55*   | 43.70*   | 0.00     | 26.15*   | 21.13*   | -0.12    | 34.95*    | 33.55*    | -0.80     | 33.33*      | 31.71*      | -1.56*      | 45.90*   | 45.90*   | 6.36*    |
| GDA_FSL1    | 17.40*   | 17.55*   | -26.15*  | 0.00     | -5.02*   | -26.27*  | 8.81*     | 7.40*     | -26.94*   | 7.19*       | 5.57*       | -27.71*     | 19.76*   | 19.76*   | -19.79*  |
| GDA_FSL4    | 22.42*   | 22.57*   | -21.13*  | 5.02*    | 0.00     | -21.25*  | 13.82*    | 12.42*    | -21.93*   | 12.20*      | 10.58*      | -22.69*     | 24.77*   | 24.77*   | -14.77*  |
| GDA_RN50    | 43.67*   | 43.82*   | 0.12     | 26.27*   | 21.25*   | 0.00     | 35.08*    | 33.67*    | -0.67     | 33.46*      | 31.83*      | -1.44       | 46.02*   | 46.02*   | 6.48*    |
| TIMO_FSL1   | 8.59*    | 8.75*    | -34.95*  | -8.81*   | -13.82*  | -35.08*  | 0.00      | -1.41     | -35.75*   | -1.62       | -3.24*      | -36.51*     | 10.95*   | 10.95*   | -28.59*  |
| TIMO_FSL4   | 10.00*   | 10.15*   | -33.55*  | -7.40*   | -12.42*  | -33.67*  | 1.41      | 0.00      | -34.34*   | -0.21       | -1.83       | -35.11*     | 12.35*   | 12.35*   | -27.19*  |
| TIMO_RN50   | 44.34*   | 44.50*   | 0.80     | 26.94*   | 21.93*   | 0.67     | 35.75*    | 34.34*    | 0.00      | 34.13*      | 32.51*      | -0.76       | 46.70*   | 46.70*   | 7.16*    |
| TIMO_S_FSL1 | 10.21*   | 10.37*   | -33.33*  | -7.19*   | -12.20*  | -33.46*  | 1.62      | 0.21      | -34.13*   | 0.00        | -1.62       | -34.89*     | 12.57*   | 12.57*   | -26.97*  |
| TIMO_S_FSL4 | 11.83*   | 11.99*   | -33.71*  | -5.57*   | -10.58*  | -31.83*  | 3.24*     | 1.83      | -32.51*   | 1.62        | 0.00        | -33.27*     | 14.19*   | 14.19*   | -25.35*  |
| TIMO_S_RN50 | 45.11*   | 45.26*   | 1.56*    | 27.71*   | 22.69*   | 4.56*    | 38.99*    | 38.59*    | 2.08      | 38.69*      | 38.65*      | 0.00        | 44.65*   | 44.83*   | 7.31*    |
| Tip_FSL1    | -2.35*   | -2.20*   | -45.90*  | -19.76*  | -24.77*  | -46.02*  | -10.95*   | -12.35*   | -46.70*   | -12.57*     | -14.19*     | -47.46*     | 0.00     | 0.00     | -39.54*  |
| Tip_FSL4    | -2.35*   | -2.20*   | -45.90*  | -19.76*  | -24.77*  | -46.02*  | -10.95*   | -12.35*   | -46.70*   | -12.57*     | -14.19*     | -47.46*     | -0.00    | 0.00     | -39.54*  |
| Tip_RN50    | 37.19*   | 37.34*   | -6.36*   | 19.79*   | 14.77*   | -6.48*   | 28.59*    | 27.19*    | -7.16*    | 26.97*      | 25.35*      | -7.92*      | 39.54*   | 39.54*   | 0.00     |

Table VIII: Statistical Comparison Matrix for 4-Shot Classification (Accuracy %). \* indicates significant differences ( $p < 0.05$ ) between models using t-test.

| Model       | APE_FSL1 | APE_FSL4 | APE_RN50 | GDA_FSL1 | GDA_FSL4 | GDA_RN50 | TIMO_FSL1 | TIMO_FSL4 | TIMO_RN50 | TIMO_S_FSL1 | TIMO_S_FSL4 | TIMO_S_RN50 | Tip_FSL1 | Tip_FSL4 | Tip_RN50 |
|-------------|----------|----------|----------|----------|----------|----------|-----------|-----------|-----------|-------------|-------------|-------------|----------|----------|----------|
| APE_FSL1    | 0.00     | -0.67    | -49.27*  | -22.54*  | -30.55*  | -50.92*  | -17.16*   | -18.81*   | -51.87*   | -18.62*     | -21.74*     | -52.66*     | 3.24*    | 3.33*    | -40.34*  |
| APE_FSL4    | 0.67     | 0.00     | -48.59*  | -21.87*  | -29.88*  | -50.24*  | -16.48*   | -18.13*   | -51.19*   | -17.95*     | -21.07*     | -51.99*     | 3.91*    | 4.01*    | -39.66*  |
| APE_RN50    | 49.27*   | 48.59*   | 0.00     | 26.73*   | 18.72*   | -1.65    | 32.11*    | 30.46*    | -2.60*    | 30.64*      | 27.52*      | -3.39*      | 52.51*   | 52.60*   | 8.93*    |
| GDA_FSL1    | 22.54*   | 21.87*   | -26.73*  | 0.00     | -8.01*   | -28.38*  | 5.38*     | 3.73*     | -29.33*   | 3.91*       | 0.80        | -30.12*     | 25.78*   | 25.87*   | -17.80*  |
| GDA_FSL4    | 30.55*   | 29.88*   | -18.72*  | 8.01*    | 0.00     | -20.37*  | 13.39*    | 11.74*    | -21.31*   | 11.93*      | 8.81*       | -22.11*     | 33.79*   | 33.88*   | -9.79*   |
| GDA_RN50    | 50.92*   | 50.24*   | 1.65     | 28.38*   | 20.37*   | 0.00     | 33.76*    | 32.11*    | -0.95     | 32.29*      | 29.17*      | -1.74       | 54.16*   | 54.25*   | 10.58*   |
| TIMO_FSL1   | 17.16*   | 16.48*   | -32.11*  | -5.38*   | -13.39*  | -34.76*  | 0.00      | -1.65     | -34.71*   | -1.47       | -4.59*      | -35.50*     | 20.40*   | 20.49*   | -23.18*  |
| TIMO_FSL4   | 18.81*   | 18.13*   | -30.46*  | -3.73*   | -11.74*  | -32.11*  | 1.65      | 0.00      | -33.06*   | 0.18        | -2.94*      | -33.85*     | 22.05*   | 22.14*   | -21.53*  |
| TIMO_RN50   | 51.87*   | 51.19*   | 2.60*    | 29.33*   | 21.31*   | 0.95     | 34.71*    | 33.06*    | 0.00      | 33.24*      | 30.12*      | -0.80       | 55.11*   | 55.20*   | 11.53*   |
| TIMO_S_FSL1 | 18.62*   | 17.95*   | -30.64*  | -3.91*   | -11.93*  | -32.29*  | 1.47      | -0.18     | -34.13*   | 0.00        | -3.12*      | -34.04*     | 21.87*   | 21.96*   | -21.71*  |
| TIMO_S_FSL4 | 21.74*   | 21.07*   | -27.52*  | -0.80    | -8.81*   | -29.17*  | 4.59*     | 2.94*     | -30.12*   | 3.12        | 0.00        | -30.92*     | 24.98*   | 25.08*   | -18.59*  |
| TIMO_S_RN50 | 52.66*   | 51.99*   | 3.39*    | 30.12*   | 22.11*   | 1.74     | 35.50*    | 33.85*    | 0.76      | 34.04*      | 30.92*      | 0.00        | 55.90*   | 55.99*   | 12.32*   |
| Tip_FSL1    | -3.24*   | -3.91*   | -52.51*  | -25.78*  | -33.79*  | -54.16*  | -10.95*   | -12.35*   | -55.11*   | -21.87*     | -24.98*     | -55.90*     | 0.00     | 0.09     | -43.58*  |
| Tip_FSL4    | -3.33*   | -4.01*   | -52.60*  | -25.87*  | -33.88*  | -54.25*  | -10.49*   | -12.14*   | -55.20*   | -21.96*     | -25.08*     | -55.99*     | -0.09    | 0.00     | -43.67*  |
| Tip_RN50    | 40.34*   | 39.66*   | -8.93*   | 17.80*   | 9.79*    | -10.58*  | 23.18*    | 21.53*    | -11.53*   | 21.71*      | 18.59*      | -12.32*     | 43.58*   | 43.67*   | 0.00     |

Table IX: Statistical Comparison Matrix for 8-Shot Classification (Accuracy %). \* indicates significant differences ( $p < 0.05$ ) between models using t-test.

| Model       | APE_FSL1 | APE_FSL4 | APE_RN50 | GDA_FSL1 | GDA_FSL4 | GDA_RN50 | TIMO_FSL1 | TIMO_FSL4 | TIMO_RN50 |
|-------------|----------|----------|----------|----------|----------|----------|-----------|-----------|-----------|
| APE_FSL1    | 0.00     | -2.17*   | -52.75*  | -28.90*  | -41.62*  | -58.26*  | -25.29*   | -30.86*   | -57.74*   |
| APE_FSL4    | 2.17*    | 0.00     | -50.58*  | -26.73*  | -39.45*  | -56.09*  | -23.12*   | -30.86*   | -57.74*   |
| APE_RN50    | 52.75*   | 50.58*   | 0.00     | 23.85*   | 11.13*   | -5.50*   | 27.46*    | 10.09     | 20.00     |
| GDA_FSL1    | 34.46*   | 33.21*   | -20.98*  | 0.00     | -12.17*  | -31.87*  | 4.19*     | 2.08      | 32.17*    |
| GDA_FSL4    | 46.64*   | 45.38*   | -8.81*   | 12.17*   | 0.00     | -19.69*  | 16.36*    | 10.09     | 20.00     |
| GDA_RN50    | 66.33*   | 65.08*   | 10.89*   | 31.87*   | 19.69*   | 0.00     | 36.06*    | 16.36*    | 10.09     |
| TIMO_FSL1   | 25.29*   | 23.12*   | -27.46*  | -3.61*   | -16.33*  | -32.97*  | 0.00      | -7.74*    | -36.62*   |
| TIMO_FSL4   | 33.03*   | 30.86*   | -19.72*  | 4.13*    | -8.59*   | -25.23*  | 7.74*     | 0.00      | -34.62*   |
| TIMO_RN50   | 59.91*   | 57.74*   | 7.16*    | 31.01*   | 18.29*   | 1.65     | 34.62*    | 36.62*    | 0.00      |
| TIMO_S_FSL1 | 26.82*   | 24.65*   | -25.93*  | -2.08    | -14.80*  | -31.44*  | 1.53      | -6.21*    | -33.03*   |
| TIMO_S_FSL4 | 35.54*   | 33.36*   | -17.22*  | 6.64*    | -6.09*   | -22.72*  | 10.24*    | 2.51      | -24.37*   |
| TIMO_S_RN50 | 59.72*   | 57.55*   | 6.97*    | 30.83*   | 18.10*   | 1.47     | 34.43*    | 36.70*    | 0.09      |
| Tip_FSL1    | -4.10*   | -5.35*   | -59.54*  | -38.56*  | -50.73*  | -70.43*  | -34.37*   | -40.64*   | -70.73*   |
| Tip_FSL4    | -4.95*   | -6.21*   | -60.40*  | -39.42*  | -51.59*  | -71.28*  | -35.23*   | -41.50*   | -71.59*   |
| Tip_RN50    | 46.18*   | 44.92*   | -9.27*   | 11.71*   | -0.46    | -20.15*  | 15.90*    | 9.63*     | -20.46*   |

Table X: Statistical Comparison Matrix for 16-Shot Classification (Accuracy %). \* indicates significant differences ( $p < 0.05$ ) between models using t-test.

| Model       | APE_FSL1 | APE_FSL4 | APE_RN50 | GDA_FSL1 | GDA_FSL4 | GDA_RN50 | TIMO_FSL1 | TIMO_FSL4 | TIMO_RN50 |
|-------------|----------|----------|----------|----------|----------|----------|-----------|-----------|-----------|
| APE_FSL1    | 0.00     | -1.25    | -55.44*  | -34.46*  | -46.64*  | -66.33*  | -30.28*   | -29.02*   | -55.44*   |
| APE_FSL4    | 1.25     | 0.00     | -54.19*  | -33.21*  | -45.38*  | -65.08*  | 25.17*    | 18.90*    | -54.19*   |
| APE_RN50    | 55.44*   | 54.19*   | 0.00     | 20.98*   | 8.81*    | -10.89*  | 36.06*    | 29.79*    | 0.00      |
| GDA_FSL1    | 34.46*   | 33.21*   | -20.98*  | 0.00     | -12.17*  | -31.87*  | 4.19*     | 2.08      | 32.17*    |
| GDA_FSL4    | 46.64*   | 45.38*   | -8.81*   | 12.17*   | 0.00     | -19.69*  | 16.36*    | 10.09     | 20.00     |
| GDA_RN50    | 66.33*   | 65.08*   | 10.89*   | 31.87*   | 19.69*   | 0.00     | 36.06*    | 16.36*    | 10.09     |
| TIMO_FSL1   | 25.29*   | 23.12*   | -27.46*  | -3.61*   | -16.33*  | -32.97*  | 0.00      | -7.74*    | -36.62*   |
| TIMO_FSL4   | 33.03*   | 30.86*   | -19.72*  | 4.13*    | -8.59*   | -25.23*  | 7.74*     | 0.00      | -34.62*   |
| TIMO_RN50   | 59.91*   | 57.74*   | 7.16*    | 31.01*   | 18.29*   | 1.65     | 34.62*    | 36.62*    | 0.00      |
| TIMO_S_FSL1 | 26.82*   | 24.65*   | -25.93*  | -2.08    | -14.80*  | -31.44*  | 1.53      | -6.21*    | -33.03*   |
| TIMO_S_FSL4 | 35.54*   | 33.36*   | -17.22*  | 6.64*    | -6.09*   | -22.72*  | 10.24*    | 2.51      | -24.37*   |
| TIMO_S_RN50 | 59.72*   | 57.55*   | 6.97*    | 30.83*   | 18.10*   | 1.47     | 34.43*    | 36.70*    | 0.09      |
| Tip_FSL1    | -4.10*   | -5.35*   | -59.54*  | -38.56*  | -50.73*  | -70.43*  | -34.37*   | -40.64*   | -70.73*   |
| Tip_FSL4    | -4.95*   | -6.21*   | -60.40*  | -39.42*  | -51.59*  | -71.28*  | -35.23*   | -41.50*   | -71.59*   |
| Tip_RN50    | 46.18*   | 44.92*   | -9.27*   | 11.71*   | -0.46    | -20.15*  | 15.90*    | 9.63*     | -20.46*   |

## D. Discussion

Our comprehensive evaluation addresses two primary research questions: (1) the comparative effectiveness of different backbones for few-shot artist classification, and (2) whether explicit style integration through StyCLIP improves performance over standard CLIP approaches.

1) *Backbone Comparison Analysis:* The experimental results demonstrate a clear hierarchy in backbone performance across all shot configurations. The standard CLIP RN50 backbone consistently outperforms both StyCLIP variants (FSL1 and FSL4) by substantial margins.

In 1-shot scenarios, models using RN50 backbone achieve accuracy ranges of 38.65%-45.96%, while StyCLIP FSL1 variants reach only 1.31%-13.49% and StyCLIP FSL4 variants achieve 1.13%-16.33%. This performance gap persists and even widens in higher-shot configurations, with RN50-based models reaching 52.91%-73.46% accuracy in 16-shot scenarios, compared to 2.63%-41.19% for StyCLIP FSL1 and 1.77%-53.36% for StyCLIP FSL4.

The statistical analysis confirms these differences are highly significant ( $p < 0.05$ ) across virtually all model-backbone combinations. This suggests that the pre-trained representations in standard CLIP RN50 are substantially more effective for artist classification than the style-augmented features learned by StyCLIP variants.

2) *Standard CLIP vs. StyCLIP Integration Analysis:* When examining the core research hypothesis whether explicit style integration improves few-shot artist classification, the results provide a definitive negative answer. Focusing on the best-performing few-shot method (TIMO/TIMO\_S) across different backbones:

### RN50 Backbone Performance:

- TIMO (standard): 43.88% (1-shot) 73.36% (16-shot)

- TIMO\_S (style-aware): 45.96% (1-shot) 73.46% (16-shot)

### StyCLIP Backbone Performance:

- TIMO with FSL1: 6.97% (1-shot) 37.00% (16-shot)
- TIMO\_S with FSL1: 7.28% (1-shot) 39.20% (16-shot)
- TIMO with FSL4: 7.37% (1-shot) 43.27% (16-shot)
- TIMO\_S with FSL4: 7.31% (1-shot) 48.75% (16-shot)

The results reveal two critical findings. First, when using the standard RN50 backbone, TIMO\_S (which includes style-aware components) shows marginal improvements over standard TIMO, but these differences are not statistically significant. Second, the StyCLIP variants consistently underperform regardless of whether style-aware components are included in the few-shot learning method. This pattern suggests that explicit Gram matrix-based style modeling disrupts rather than enhances pre-trained semantic representations. While CLIP's pre-trained features appear to already encode sufficient stylistic information for artist classification, the architectural modifications in StyCLIP interfere with this implicit knowledge.

Standard CLIP with RN50 backbones consistently outperforms both StyCLIP variants across all shot configurations. In 16-shot scenarios, standard CLIP achieves 73.46% accuracy while the best StyCLIP variant reaches only 48.75%, a statistically significant 25 percentage point difference. Among few-shot methods, TIMO and TIMO\_S demonstrate superior performance, with no significant differences between them when using standard RN50.

Classification accuracy improves substantially with more support examples (TIMO\_S: 45.96% to 73.46% from 1-shot to 16-shot). However, both StyCLIP variants consistently underperform across all conditions, suggesting that explicit Gram matrix-based style modeling may interfere with pre-trained semantic representations rather than enhance them.

The experimental results reveal several key insights across different shot configurations. In 1-shot scenarios with extremely limited data, CLIP-based models with RN50 backbone (particularly TIMO and TIMO\_S) achieve significantly superior results compared to StyCLIP variants and other methods. The performance of approaches that explicitly integrate stylistic components remains substantially lower, suggesting that adding these features does not aid generalization under minimal data conditions.

As the number of support examples increases to 2-shot and 4-shot, all models show performance improvements, but the gap between standard CLIP RN50 and StyCLIP versions remains pronounced. Standard models continue to dominate, while solutions with custom backbones struggle to exceed 25% accuracy in lower-shot scenarios.

In higher-shot configurations (8-shot and 16-shot), while performance differences between models decrease slightly, CLIP RN50 and TIMO\_S remain the most effective, achieving over 73% accuracy in 16-shot cases. StyCLIP versions fail to bridge this performance gap, confirming that explicit integration of style information through Gram matrices does not provide concrete benefits for few-shot artist classification.

## VI. CONCLUSION

This work investigated the integration of explicit style representations into CLIP architectures for few-shot artist classification through our proposed StyCLIP approach. Our comprehensive experimental evaluation across multiple shot configurations (1, 2, 4, 8, and 16-shot) provides clear answers to our research objectives.

**Backbone Effectiveness:** The standard CLIP RN50 backbone demonstrates superior performance compared to both StyCLIP variants (FSL1 and FSL4) across all experimental conditions. The performance gap is substantial, with RN50-based models achieving 38.65%-73.46% accuracy compared to 1.13%-53.36% for StyCLIP variants, indicating that pre-trained CLIP representations are more effective for artistic classification than explicitly style-augmented features.

**Style Integration Impact:** Contrary to our initial hypothesis, explicit integration of Gram matrix-based style features does not improve few-shot artist classification performance. When comparing TIMO (standard) vs. TIMO\_S (style-aware) using RN50 backbone, differences are minimal and statistically insignificant, suggesting that CLIP’s pre-trained features already capture sufficient stylistic information for artist identification.

**Architectural Implications:** The consistent underperformance of StyCLIP backbones indicates that architectural modifications designed to explicitly model artistic style may disrupt the semantic-visual alignment learned during CLIP’s pre-training phase. This disruption outweighs any potential benefits from explicit style modeling.

Our findings suggest that foundation models like CLIP inherently encode rich stylistic information through their large-scale pre-training, making additional explicit style modeling redundant or potentially counterproductive. Future research should explore alternative approaches such as attention-based style integration, prompt

[3] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, “Learning transferable visual models from natural language supervision,” in *International conference on machine learning*. PmLR, 2021, pp. 8748–8763.

engineering techniques, or domain-specific fine-tuning strategies that preserve the integrity of pre-trained representations while enhancing artistic understanding.

The negative results provide valuable insights into the complexity of augmenting foundation models and highlight the importance of carefully evaluating architectural modifications against strong pre-trained baselines in specialized domains.

## VII. ACKNOWLEDGMENTS

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## REFERENCES

- [1] L. A. Gatys, A. S. Ecker, and M. Bethge, “Image style transfer using convolutional neural networks,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2016.
- [2] J. Snell, K. Swersky, and R. Zemel, “Prototypical networks for few-shot learning,” in *Advances in Neural Information Processing Systems*, I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds., vol. 30. Curran Associates, Inc., 2017. [Online]. Available: [https://proceedings.neurips.cc/paper\\_files/paper/2017/file/cb8da6767461f2812ae4290eac7cbc42-Paper.pdf](https://proceedings.neurips.cc/paper_files/paper/2017/file/cb8da6767461f2812ae4290eac7cbc42-Paper.pdf)
- [4] R. Zhang, W. Zhang, R. Fang, P. Gao, K. Li, J. Dai, Y. Qiao, and H. Li, “Tip-adapter: Training-free adaption of clip for few-shot classification,” in *European conference on computer vision*. Springer, 2022, pp. 493–510.
- [5] X. Zhu, R. Zhang, B. He, A. Zhou, D. Wang, B. Zhao, and P. Gao, “Not all features matter: Enhancing few-shot clip with adaptive prior refinement,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 2605–2615.
- [6] Z. Wang, J. Liang, L. Sheng, R. He, Z. Wang, and T. Tan, “A hard-to-beat baseline for training-free clip-based adaptation,” *arXiv preprint arXiv:2402.04087*, 2024.
- [7] Y. Li, J. Guo, L. Qi, W. Li, and Y. Shi, “Text and image are mutually beneficial: Enhancing training-free few-shot classification with clip,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 39, no. 5, 2025, pp. 5039–5047.
- [8] G. Castellano, V. Digeno, G. Sansaro, and G. Vessio, “Artgraph,” Mar. 2022. [Online]. Available: <https://doi.org/10.5281/zenodo.6337958>
- [9] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, “Optuna: A next-generation hyperparameter optimization framework,” in *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*, 2019, pp. 2623–2631.